

Module 16) Django Rest Framework

Section A: Conceptual Understanding

1. Why are REST APIs preferred for mobile and frontend-heavy applications?

Ans:- REST APIs are preferred because they allow different applications, such as mobile apps and frontend frameworks, to communicate with the backend efficiently. They exchange data in JSON format, which is lightweight, fast, and easy to process. REST APIs are platform-independent, making them suitable for web, mobile, and desktop applications.

2. How do serializers act as a validation layer and how can custom field-level validation be implemented?

Ans:- Serializers convert complex data such as Django model instances into JSON format and vice versa. They also validate incoming data before it is saved to the database. Custom field-level validation can be implemented by creating methods such as `validate_fieldname()` inside the serializer. This helps ensure that only valid data is accepted.

3. Why is it important to use appropriate HTTP status codes (201 Created vs 200 OK)?

Ans:- HTTP status codes help clients understand the result of an API request. A **200 OK** status indicates that a request was successful and data was retrieved or updated successfully. A **201 Created** status indicates that a new resource has been successfully created. Using the correct status codes improves API clarity and follows REST standards.

4. Why is pagination required when listing doctors and how does it impact database performance?

Ans:- Pagination is used to limit the number of records returned in a single request. If thousands of doctor records exist, returning all of them at once can slow down the API and increase server load. Pagination improves performance,

reduces memory usage, and provides a better user experience by returning data in smaller chunks.

5. What are the benefits of using ViewSets over APIView for rapid CRUD development?

Ans:- ViewSets reduce the amount of code required for CRUD operations. Instead of writing separate methods for GET, POST, PUT, and DELETE requests, a `ModelViewSet` automatically provides these operations. This speeds up development, improves code readability, and follows Django REST Framework best practices.

6. Why should `transaction.atomic()` be used when creating related doctor records?

Ans:- The `transaction.atomic()` method ensures that database operations are completed as a single transaction. If an error occurs during the process, all changes are rolled back automatically. This prevents partial or inconsistent data from being stored in the database and helps maintain data integrity.